VIBE READER

Research Abstracts & Objectives

Project Research Scope

Vibe Reader combines: digital document reading, text-to-speech, expressive/emotion-aware narration, synchronized audio-text reading, mood-aware personalization, distraction-free reading, and personalized audiobook/content recommendations.

Research Area Coverage

Research Area	Relevant Sources	Vibe Reader Application
Expressive TTS	1, 5	Expressive and emotion-aware narration
TTS & comprehension	2, 3, 4	Read + Listen and synchronized reading
Personalized affective computing	7, 8, 17	Mood-aware personalization and adaptive UI
Digital distraction	9, 10, 11, 12, 16	Zen / Focus Mode
E-book UX	13, 14, 15	Reader interface, navigation and comfort
Recommendations	6	Personalized audiobook/content discovery

1. Computational Narrative Understanding for Expressive Text-to-Speech

Year: 2026 | **Authors:** Gaspard Michel, Elena V. Epure, Christophe Cerisara

Objective

To improve expressive text-to-speech by using narrative and contextual information from audiobook speech, particularly character dialogue, to guide more natural and intelligible expressive delivery.

Crisp Abstract Synthesis

The study addresses the limited use of audiobook data for expressive TTS. It introduces LibriQuote, a 5.3K-hour corpus of expressive character quotations enriched with contextual pseudo-labels describing intended delivery. Experiments show that fine-tuning a flow-matching model improves expressivity and intelligibility, while autoregressive TTS trained from scratch can also gain expressive capability. The work provides data, code, and evaluation resources for reproducible research.

Relevance to Vibe Reader

Direct foundation for expressive, character-aware, and narrative-aware audiobook narration in Vibe Reader.

Source

https://aclanthology.org/2026.findings-acl.308/

2. The Effects of Text-to-Speech on Reading Comprehension in Junior High School Students with Reading Disabilities: An Eye-Tracking Study in Taiwan

Year: 2026 | **Authors:** Ching-Hsuan Chen, Li-Yu Hung, Yu-Cin Jian

Objective

To examine how TTS affects reading comprehension and eye-movement behavior across students with dyslexia, ADHD-related reading difficulties, and typical development.

Crisp Abstract Synthesis

The study compares silent reading and TTS-supported reading using a mixed factorial design and eye-tracking. TTS significantly improved comprehension for students with ADHD-related reading difficulties, while students with dyslexia showed reduced reading time and cognitive effort without a comprehension gain. TTS also attenuated differences in eye-movement patterns between groups. The findings indicate that TTS benefits depend on the reader's cognitive and reading profile, supporting adaptive rather than one-size-fits-all reading assistance.

Relevance to Vibe Reader

Supports adaptive Read + Listen functionality and user-specific TTS behavior.

Source

https://link.springer.com/article/10.1007/s11145-025-10738-5

3. Impact of Text-to-Speech Features on the Reading Comprehension of Children with Reading and Language Difficulties

Year: 2023 | **Authors:** Jennifer L. Keelor et al.

Objective

To determine whether different TTS configurations, including synchronized highlighting, improve reading comprehension in children with reading and language difficulties.

Crisp Abstract Synthesis

Children aged 8–12 completed reading-comprehension tasks under silent reading, read-aloud, listen-only, TTS without highlighting, and TTS with highlighting conditions. TTS significantly improved comprehension compared with no TTS, with both highlighted and non-highlighted TTS producing higher comprehension than silent reading. The results provide evidence for combining auditory support with visual text presentation in assistive reading systems.

Relevance to Vibe Reader

Directly supports synchronized audio-text reading and highlighting in Vibe Reader.

Source

https://doi.org/10.1007/s11881-023-00281-9

4. Does Use of Text-to-Speech and Related Read-Aloud Tools Improve Reading Comprehension for Students with Reading Disabilities? A Meta-Analysis

Year: 2017 | **Authors:** Wood et al.

Objective

To synthesize evidence on whether TTS and related read-aloud technologies improve reading comprehension for students with reading difficulties.

Crisp Abstract Synthesis

This meta-analysis evaluates research on TTS and read-aloud tools for students with reading difficulties. The pooled results indicate a positive overall effect on reading comprehension, while also showing that study design and other moderating factors influence effectiveness. The authors conclude that TTS can assist comprehension but emphasize the need for further research on when and for whom these tools are most effective.

Relevance to Vibe Reader

Provides broad research justification for TTS and accessibility-oriented Read + Listen features.

Source

https://pmc.ncbi.nlm.nih.gov/articles/PMC5494021/

5. Expressive Speech Synthesis in MARY TTS Using Audiobook Data and EmotionML

Year: 2013 | **Authors:** Marcela Charfuelan Oliva, Ingmar Steiner

Objective

To develop an expressive speech-synthesis framework that represents and controls vocal expressivity using audiobook data and EmotionML.

Crisp Abstract Synthesis

The paper presents a MARY TTS framework that combines audiobook data, expressive voice styles, and EmotionML for controllable expressive speech synthesis. Audiobook sentences are grouped according to voice style using acoustic-feature analysis, and EmotionML is integrated to represent discrete emotions and emotion dimensions. Preliminary perceptual results demonstrate the feasibility of controlling different expressive voice styles.

Relevance to Vibe Reader

Technical precedent for emotion-controlled narration and expressive voice generation.

Source

https://www.dfki.de/en/web/research/projects-and-publications/publication/6840

6. Personalized Audiobook Recommendations at Spotify Through Graph Neural Networks

Year: 2024 | **Authors:** Marco De Nadai et al.

Objective

To build a scalable personalized audiobook recommendation system that can operate under sparse audiobook interaction data and large-scale production constraints.

Crisp Abstract Synthesis

The work addresses the difficulty of recommending audiobooks when users have limited interaction history and audiobooks cannot be easily sampled before purchase. It introduces 2T-HGNN, combining heterogeneous graph neural networks with a two-tower architecture to model content relationships while maintaining low latency and scalability. Large-scale evaluation reports substantial gains in audiobook starts and streaming, demonstrating the value of graph-based personalization.

Relevance to Vibe Reader

Supports personalized audiobook/content recommendation in Vibe Reader.

Source

https://arxiv.org/abs/2403.05185

7. Beyond One-Size-Fits-All: A Survey of Personalized Affective Computing in Human-Agent Interaction

Year: 2023 | **Authors:** Jialin Li, Maha Elgarf, Alia Waleed, Hanan Salam

Objective

To survey personalization strategies in affective computing and establish a framework for adapting emotion-aware systems to individual users or user groups.

Crisp Abstract Synthesis

The survey examines personalized affective-computing approaches across interaction modes and contexts. It organizes methods into data-level and model-level approaches, including user-specific models, group-specific models, weighting, feature augmentation, generative methods, fine-tuning, and multitask learning. The authors formulate personalization problems and analyze trends across affective-computing tasks, contexts, and levels of personalization, providing a roadmap for future research.

Relevance to Vibe Reader

Provides the conceptual basis for mood-aware personalization and adaptive interaction.

Source

https://arxiv.org/abs/2304.00377

8. Adaptive User Interface Design and Analysis Using Emotion Recognition Through Facial Expressions and Body Posture from an RGB-D Sensor

Year: 2020 | **Authors:** Selma Medjden, Naveed Ahmed, Mohammed Lataifeh

Objective

To design and evaluate an adaptive user interface that automatically changes according to recognized user emotional states and physiological cues.

Crisp Abstract Synthesis

The study develops an adaptive UI using real-time facial-expression and body-posture recognition from an RGB-D/Kinect sensor. Six basic emotions and physiological state are recognized, with a reported recognition accuracy above 90%. The detected state drives automatic interface adaptation, which is evaluated against manual and hybrid adaptation. The results indicate that hybrid adaptation can improve usability, productivity, efficiency, and user experience.

Relevance to Vibe Reader

Supports emotion-aware themes, adaptive UI behavior, and mood-responsive reading interfaces.

Source

https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0235908

9. Effects of Digital Reading With On-Screen Distractions: An Eye-Tracking Study

Year: 2025 | **Authors:** Angelica Ronconi, Lucia Mason, Lucia Manzione, Anne Schüler

Objective

To examine how common on-screen distractions such as advertisements and social-media notifications affect text processing, cognitive load, and comprehension during digital reading.

Crisp Abstract Synthesis

University students read digital science texts with and without simple on-screen distractions while their eye movements were recorded. The study found that participants spent little time looking at the distractions and that these passive distractions did not significantly affect immediate text processing, perceived cognitive load, or comprehension. The findings highlight the role of readers' top-down attentional control and distinguish passive visual distractions from active multitasking.

Relevance to Vibe Reader

Provides nuanced evidence for designing distraction controls and explains when passive distractions may or may not require intervention.

Source

https://doi.org/10.1111/jcal.13106

10. Distractions in Digital Reading: A Meta-Analysis of Attentional Interference Effects

Year: 2025 | **Authors:** Yamin Shen

Objective

To quantify the overall effect of attentional distractions on comprehension in digital reading and identify factors that moderate this effect.

Crisp Abstract Synthesis

The meta-analysis synthesizes 32 empirical studies of attentional distraction in digital reading environments. The overall effect is negative, with Hedges' g reported as -0.6411, indicating substantial interference with reading comprehension. Moderator analyses consider distraction type, study design, reader age, educational level, device, effort, time control, and article characteristics. The findings strengthen the case for reducing disruptive attentional interference in digital reading systems.

Relevance to Vibe Reader

Strong evidence base for Zen Mode, Focus Mode, notification suppression, and distraction-aware design.

Source

https://pubmed.ncbi.nlm.nih.gov/41368163/

11. Digital Distractions in Reading: Investigating Impact of Cognitive Control Training on Reading Behavior and Outcomes

Year: 2024 | **Authors:** Jayasankar Santhosh, Shrishti Jagtap, Andreas Dengel, Shoya Ishimaru

Objective

To measure the effects of notifications and pop-ups on reading comprehension, reading flow, frustration, and eye movements, and to examine whether cognitive-control training can mitigate distraction effects.

Crisp Abstract Synthesis

The study evaluates digital distractions using eye-tracking and survey data across multiple documents. Notifications and pop-ups produced significant disruption to reading flow, reduced comprehension, and increased frustration. The researchers also examined cognitive-control training and observed improved comprehension in distracted digital-reading conditions. The work provides quantitative support for interface-level strategies that minimize notification and pop-up interruptions.

Relevance to Vibe Reader

Directly supports notification management and distraction-reduction mechanisms in Zen Mode.

Source

https://dl.acm.org/doi/10.1145/3675094.3677591

12. What Breaks the Flow of Reading? A Study on Characteristics of Attentional Disruption During Digital Reading

Year: 2022 | **Authors:** Guillaume Chevet, Thierry Baccino, Annie Vinter, Véronique Drai-Zerbib

Objective

To characterize attentional disruptions during digital reading in naturalistic settings and examine their relationship with reading comprehension and reader behavior.

Crisp Abstract Synthesis

The study records interruptions, their type, duration, location, mental effort, and emotional valence, together with self-caught mind wandering, during reading at home. Across 74 participants, attentional disruptions occurred approximately every four minutes and were more frequent early in the text. Media-related interruptions were more common among high media multitaskers, while pleasure during reading predicted comprehension. The findings emphasize the importance of interruption recovery and sustained reading flow.

Relevance to Vibe Reader

Supports flow-preserving interfaces, interruption management, and continuous reading experiences.

Source

https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.987964/full

13. Enhancement of Reading Experience: Users' Behavior Patterns and the Interactive Interface Design of Tablet Readers

Year: 2014 | **Authors:** Kuo-Liang Huang, Kuo-Hsiang Chen, Chun-Heng Ho

Objective

To identify differences in user behavior across tablet and other digital reading systems, derive interface requirements, and evaluate design changes intended to improve user experience and behavioral intention.

Crisp Abstract Synthesis

The research uses contextual inquiry to identify reading behaviors and interface requirements, develops design solutions, and evaluates them using the Technology Acceptance Model. The improved interface performed significantly better in perceived usefulness, perceived ease of use, user attitude, and behavioral intention. The study highlights the importance of understanding reading context and behavior when designing digital reading interfaces.

Relevance to Vibe Reader

Useful for reader UI design, interaction patterns, usability, and experience optimization.

Source

https://doi.org/10.1108/lht-01-2014-0002

14. Realistic Electronic Books

Year: 2012 | **Authors:** Veronica Liesaputra, Ian H. Witten

Objective

To design an electronic-book model that combines physical-book interaction cues with digital capabilities and evaluate whether the resulting interface improves reading-task performance and user preference.

Crisp Abstract Synthesis

The proposed electronic-book model emulates physical-book properties such as page turning, visual location cues, bookmarks, and annotations while adding digital functions including hyperlinks, multimedia, full-text search, synonym search, cross-referencing, and automatic indexing. Usability studies comparing the system with HTML, PDF, and physical books found that participants completed tasks more efficiently without losing accuracy and preferred the new interface.

Relevance to Vibe Reader

Supports advanced e-book navigation, search, annotation, and interactive reading features.

Source

https://www.sciencedirect.com/science/article/abs/pii/S1071581912000316

15. Usability Evaluation of E-books

Year: 2009 | **Authors:** Yen-Yu Kang, Mao-Jiun J. Wang, Rungtai Lin

Objective

To evaluate e-book usability by comparing electronic and conventional book reading using objective measures of reading performance and visual fatigue.

Crisp Abstract Synthesis

An experiment with 20 junior college students compared e-book and conventional-book reading using reading performance and critical flicker fusion measures. The study found higher eye fatigue during e-book reading and better reading performance for conventional books. The results demonstrate that electronic reading can impose different visual demands and that e-book usability should account for reading behavior and visual comfort.

Relevance to Vibe Reader

Supports attention to reader comfort, typography, display design, and usability in Vibe Reader.

Source

https://www.sciencedirect.com/science/article/abs/pii/S0141938208000826

16. Reading in the Age of Digital Distraction

Year: 2021/2022 | **Authors:** Ziming Liu

Objective

To investigate the extent and effects of digital distraction during screen reading, with particular emphasis on multitasking and communication-related interruptions.

Crisp Abstract Synthesis

The study examines digital distraction from the perspectives of digital content, readers, and the reading medium. Survey and analysis results indicate substantial distraction among college students, with communication-based activities identified as major sources. Many participants preferred print reading to reduce distractions and improve concentration. The work emphasizes that screen reading is strongly associated with multitasking and that repeated multitasking can impose substantial costs on academic reading.

Relevance to Vibe Reader

Provides broad motivation for distraction-free reading environments and focused reading modes.

Source

https://www.sciencedirect.com/org/science/article/abs/pii/S0022041821000011

17. Good GUIs, Bad GUIs: Affective Evaluation of Graphical User Interfaces

Year: 2024 | **Authors:** Syrine Haddad, Kayhan Latifzadeh, Saravanakumar Duraisamy, Jean Vanderdonckt, Luis A. Leiva et al.

Objective

To investigate whether physiological and behavioral signals can provide an early, implicit assessment of users' affective responses to GUI designs.

Crisp Abstract Synthesis

The work explores affective evaluation of GUI quality without relying solely on explicit user feedback. In a controlled study, 32 participants viewed good and bad GUI designs while eye tracking, facial-expression video, and EEG signals were recorded. The research demonstrates the potential of multimodal physiological signals for assessing affective responses to interface designs and for informing emotion-aware UI development.

Relevance to Vibe Reader

Supports affective UI evaluation and future validation of mood-responsive Vibe Reader themes and interfaces.

Source

https://orbilu.uni.lu/handle/10993/61604

Key Literature-Based Research Gap

The supplied literature covers expressive TTS, reading comprehension, affective computing, adaptive interfaces, digital distraction, e-book usability, and recommendation systems. These areas are generally investigated as separate problems. A potential research opportunity for Vibe Reader is to integrate them into one adaptive reading environment that can combine reader-specific personalization, expressive narration, synchronized reading/listening, content recommendation, and distraction-aware interaction.

Recommended Priority Sources for Literature Review

1. 1. Computational Narrative Understanding for Expressive Text-to-Speech
2. 2. The Effects of Text-to-Speech on Reading Comprehension in Junior High School Students with Reading Disabilities: An Eye-Tracking Study in Taiwan
3. 3. Impact of Text-to-Speech Features on the Reading Comprehension of Children with Reading and Language Difficulties
4. 10. Distractions in Digital Reading: A Meta-Analysis of Attentional Interference Effects
5. 7. Beyond One-Size-Fits-All: A Survey of Personalized Affective Computing in Human-Agent Interaction
6. 5. Expressive Speech Synthesis in MARY TTS Using Audiobook Data and EmotionML

Academic Use Note

Before submitting a final academic manuscript, verify the final bibliographic metadata (authors, publication year, venue, DOI, volume/issue, and page range) against the publisher, conference, or repository record. The source links in this document are retained for that purpose.